

Phonon Pulse Templates for SuperCDMS SNOLAB Run 4

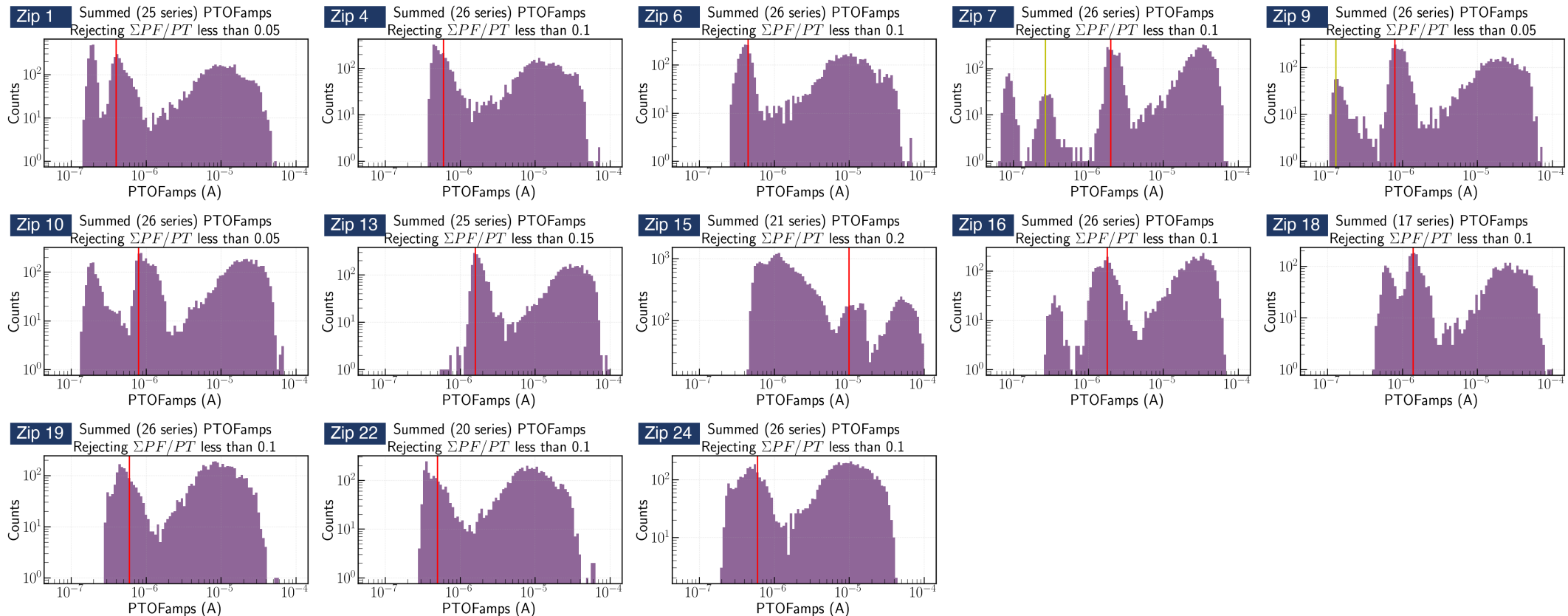
K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

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From the Ge-activation K-line to an event sample

Goal: one clean, minimally-biased pulse population per detector × channel

- Per detector, the ops-shift study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line (position $\times/\div 1.35$) — deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series → ~120 GB raw cache; every later cut stays explicit and reversible

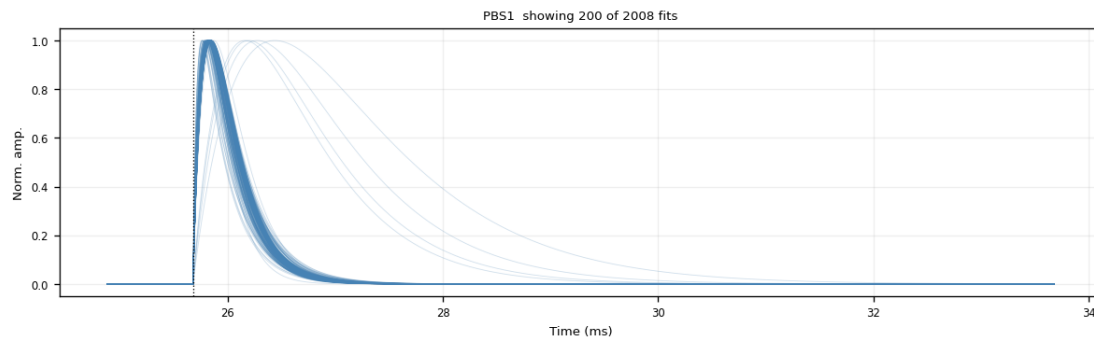


All 13 detectors, PTOFamps spectra (ops-shift study): red line = fitted mean of the 10.37 keV K-line. Selection window per detector = red-line position $\times/\div 1.35$.

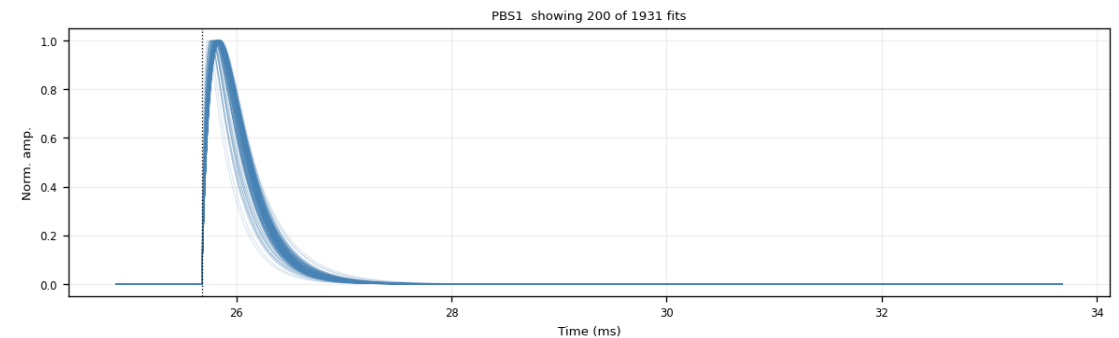
Per-trace algorithm: low-pass → free-pretrigger fit → align

- 1. 100 kHz 4th-order Butterworth low-pass (steady-state init — no filter start-up transient)
- 2. Baseline = median of samples 2000–12000, subtracted; normalize by the global trace peak
- 3. Two-exponential fit — all 5 parameters free, including the onset:
 - $y(t) = A \cdot [\exp(-(t-t_0)/\tau_{\text{fall}}) - \exp(-(t-t_0)/\tau_{\text{rise}})] + b$, $t_0 \in 16050 \pm 3000$ samples
 - fit window 12550–24050 (stride 4); $\tau_{\text{rise}} \in [1 \mu\text{s}, 5 \text{ ms}]$, $\tau_{\text{fall}} \in [10 \mu\text{s}, 20 \text{ ms}]$
- 4. Quality numbers — $\text{fit_ok} := A > 0$ and $0 < \tau_{\text{rise}} < \tau_{\text{fall}}$; $\text{NRMSE} := \text{RMS}(\text{residual})/\text{fit peak}$ (recorded, no cut yet)
- 5. Align — shift the measured trace by (fitted onset – 16050), sub-sample interpolation: a pure translation

• Why a free onset? Real trigger times vary event-by-event — fitted onsets cluster ≈ 230 samples after the nominal 16050. Pinning the onset distorts every other parameter.



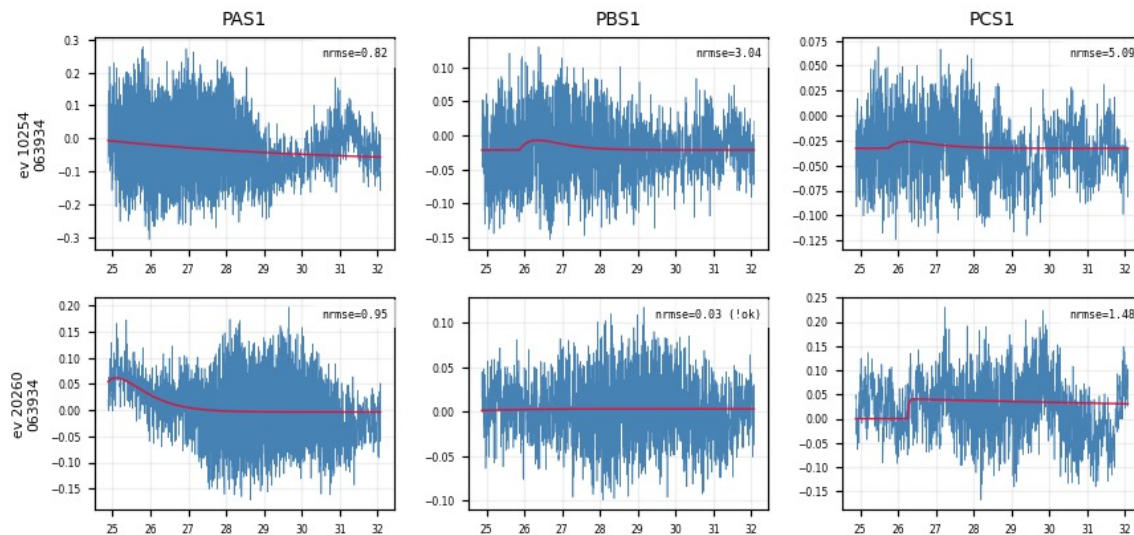
Fitted curves at common onset, peak-normalized — ALL fit_ok events (no cut)



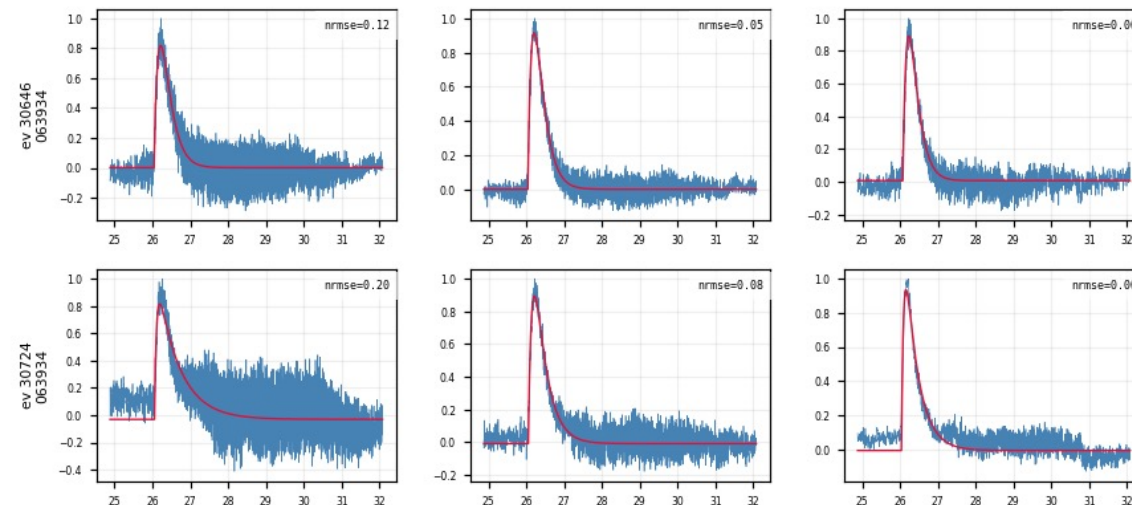
Same, after $\text{NRMSE} \leq 0.4$ — a single tight shape family remains (Z7 PBS1)

Fit quality at a glance — event × channel grids

- One row per event, one column per channel: low-passed trace (blue) vs 2-exp fit (red), NRMSE stamped per panel
- The same event fits consistently across channels; a noise trigger fails in all channels at once
- Every figure carries its full processing chain in the header → self-documenting



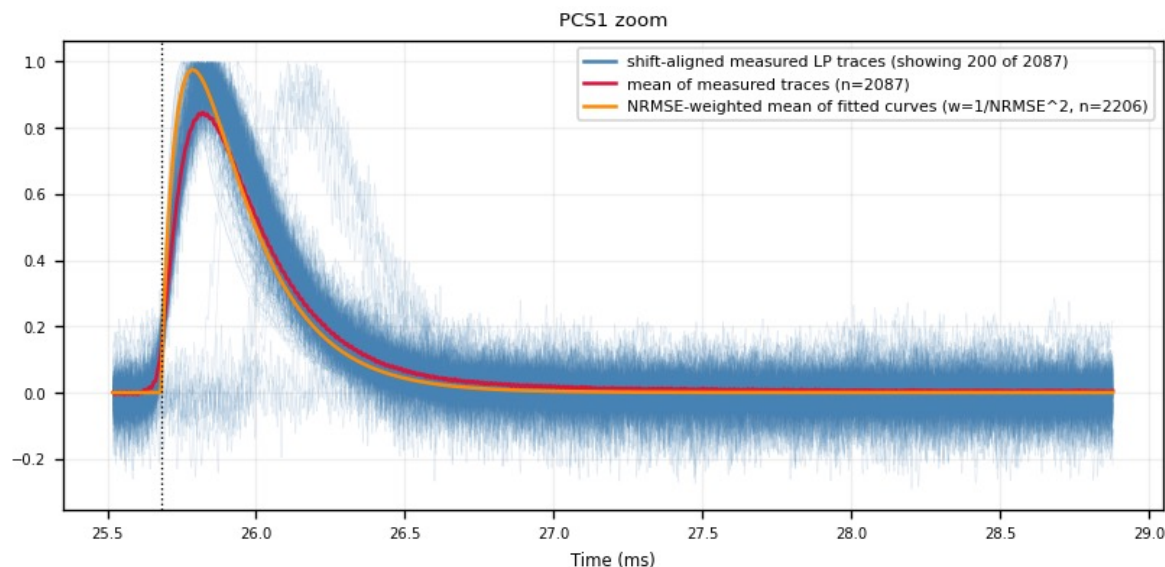
Noise triggers: the fit fails in every channel at once



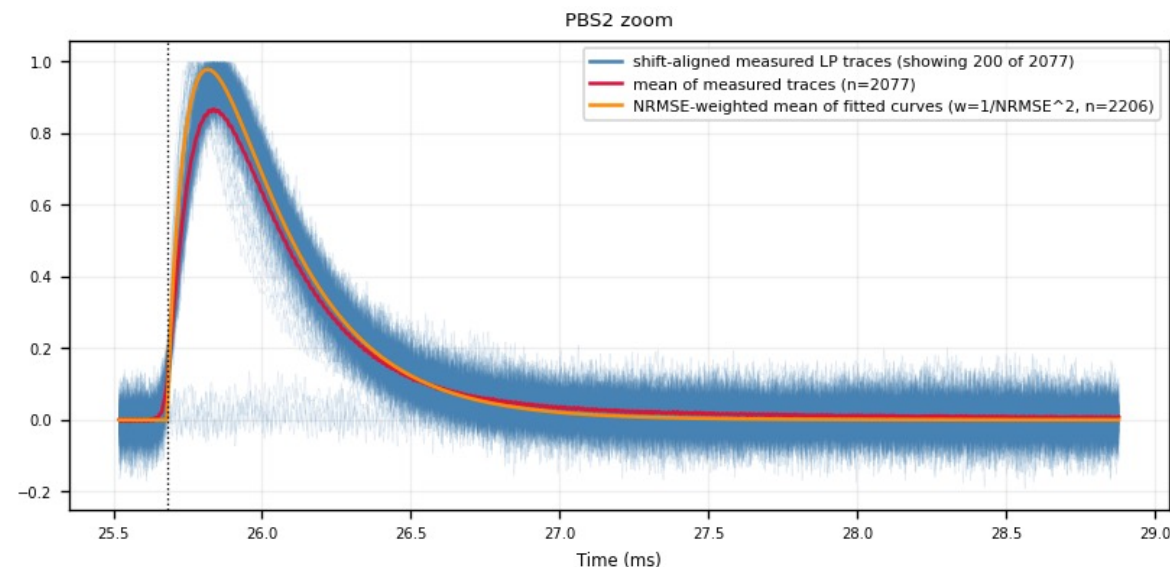
K-line events: consistent good fits across channels

Alignment result — overlaid measured traces

- All fit_ok traces shifted to the common onset (blue) + point-by-point mean (red) + NRMSE-weighted mean of the fitted curves (orange, $w = 1/\max(\text{NRMSE}, 0.01)^2$)
- Peak zoom on a quiet detector: a tight bundle — the template input is well defined. Note the faint displaced bundle beside the main one, visible on both crystal faces (S1 and S2) — more on it later



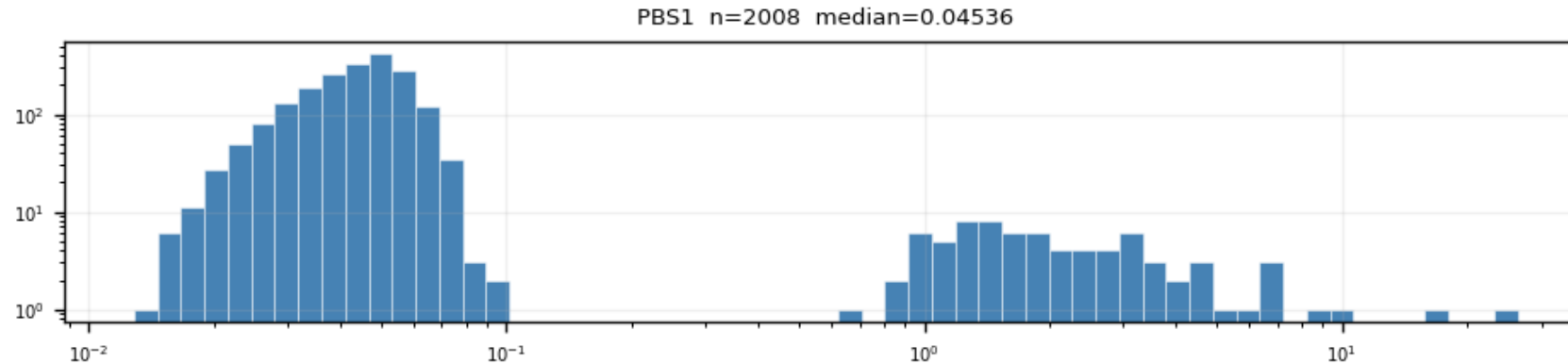
Z7 PCS1 (S1 face) — peak zoom; the faint displaced bundle is clearly visible



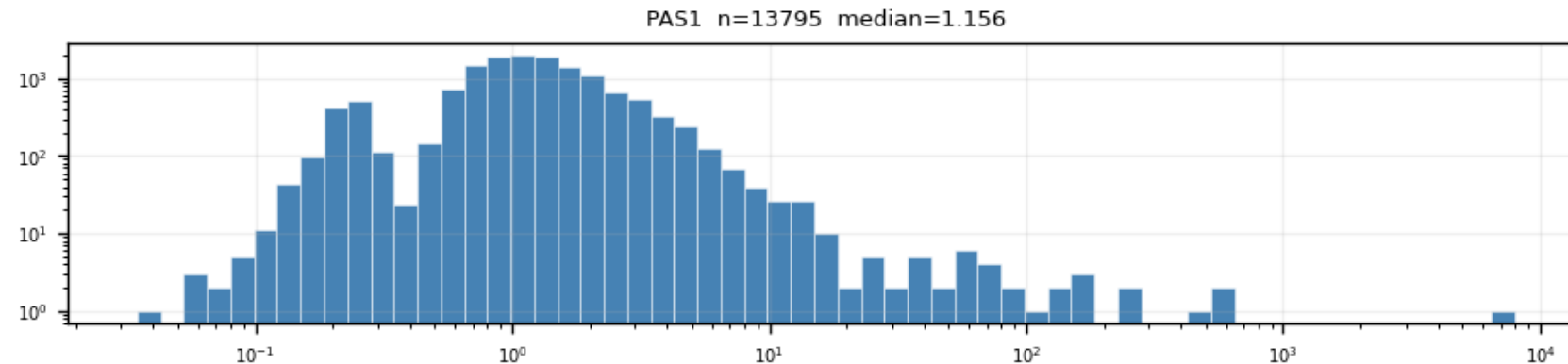
Z7 PBS2 (S2 face) — same faint displaced bundle

Quality cut 1: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05 – 0.1) vs noise triggers (≈ 1 – 2), valley at ≈ 0.4 – 0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): the PTOF window admits a noise-dominated mixture — this cut is what extracts the real-pulse population



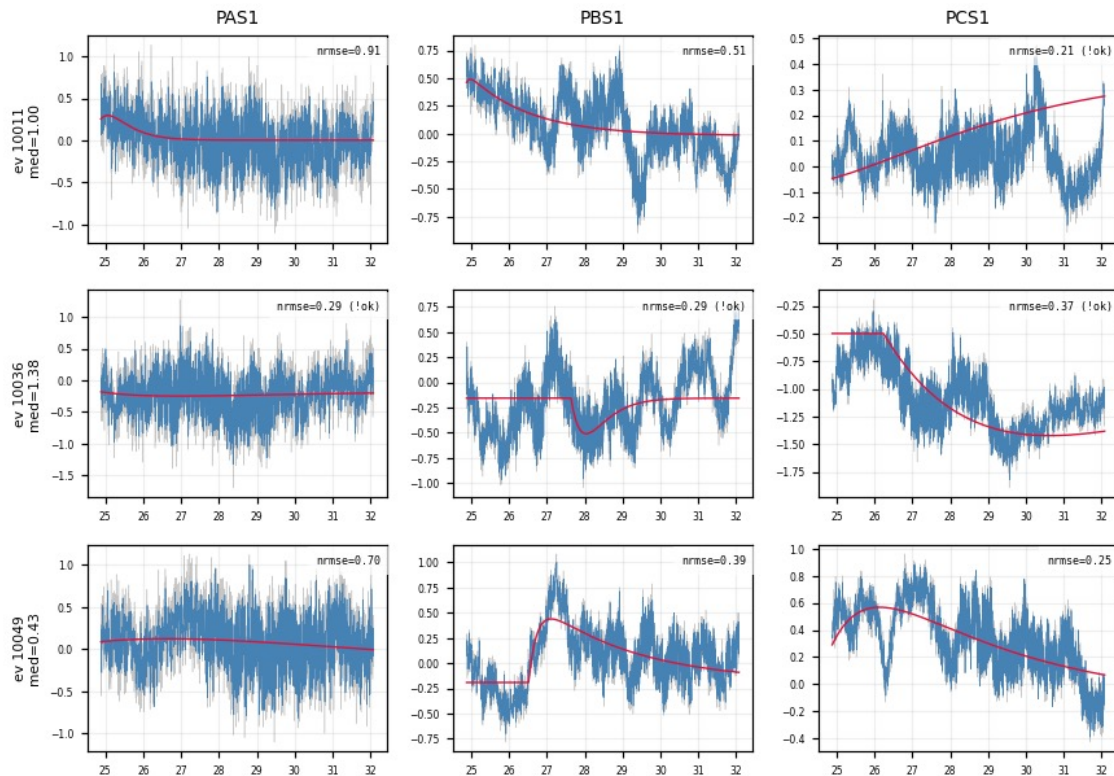
Z7 PBS1 (quiet): two clean populations, log-log axes



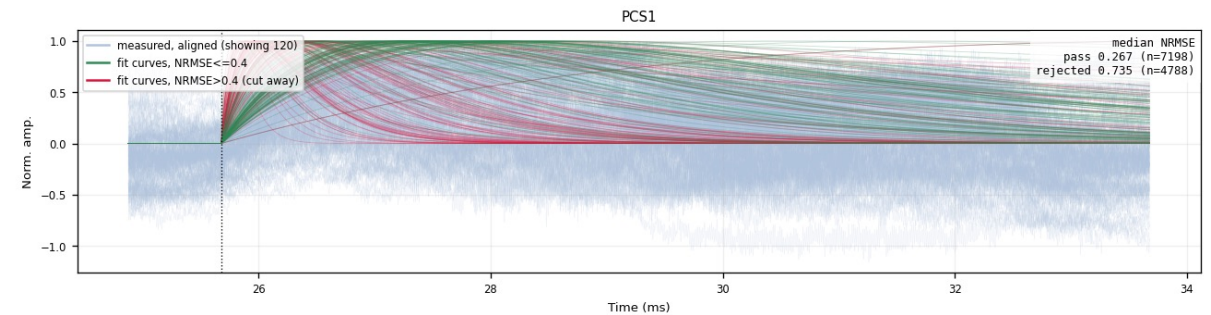
Z22 PAS1 (weak): noise population dominates — the window alone is not enough

The rejected population is noise — verified in the raw traces

- Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics
- Fan-cut view: fitted curves that pass (green) vs cut away (red) on top of the aligned data, median NRMSE of each population stamped per channel



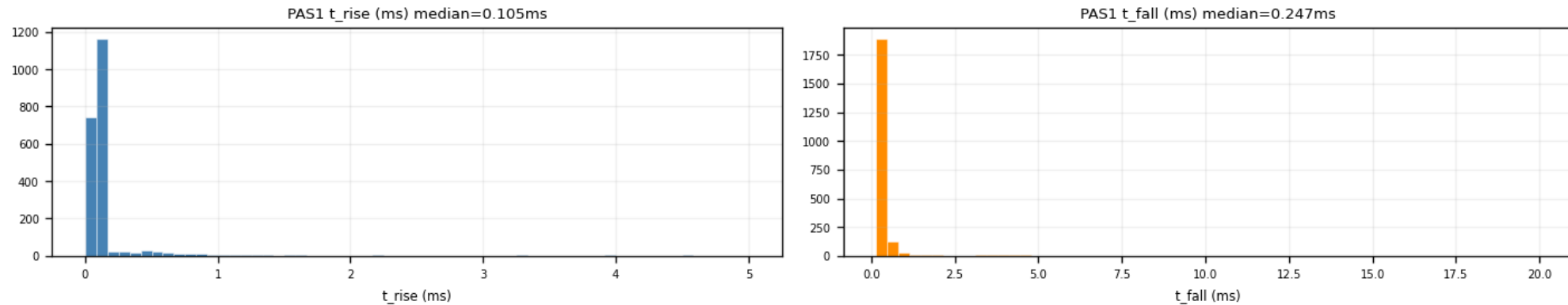
Z22: NRMSE-rejected events — raw traces are noise



Z22 PCS1: pass (green) vs rejected (red)

Quality cut 2: $\tau_{\text{rise}} \leq 0.3$ ms — against slow baseline drift

- On noisy-window detectors a smooth, slow baseline drift can survive the NRMSE cut — a slow 2-exp hugs it with a small residual
- The fast-pulse population sits at $\tau_{\text{rise}} \approx 0.1$ ms (p90 ≈ 0.15 ms) — cleanly separated from the drift tail, so a 0.3 ms ceiling removes the leakage
- Cost ≈ 2 –5% of signal on quiet detectors; known trade-off: it also trims the small population of genuine slow-rise pulses (final decision pending)

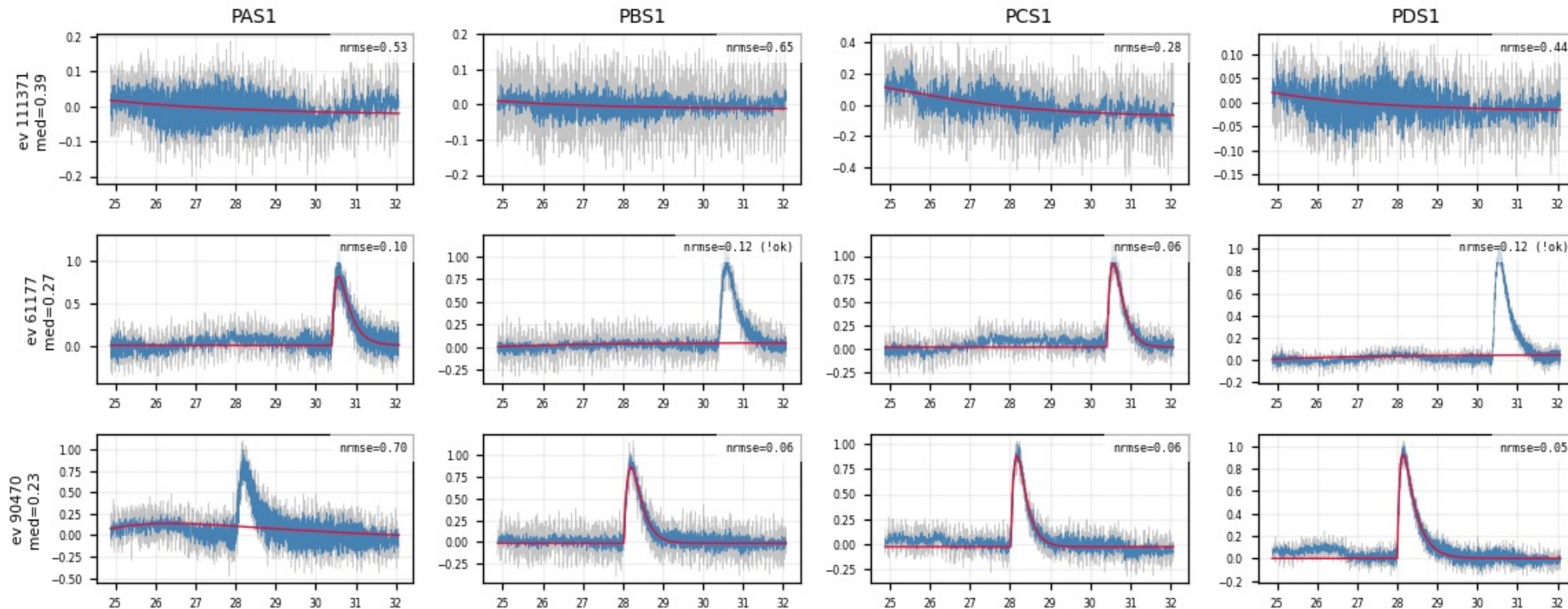


Z7 PAS1: fitted τ_{rise} (median 0.105 ms) and τ_{fall} distributions

- On quiet detectors both τ_{rise} and τ_{fall} distributions are narrow — the pulse shape is stable across events

A genuine slow-pulse population (“shadow” events)

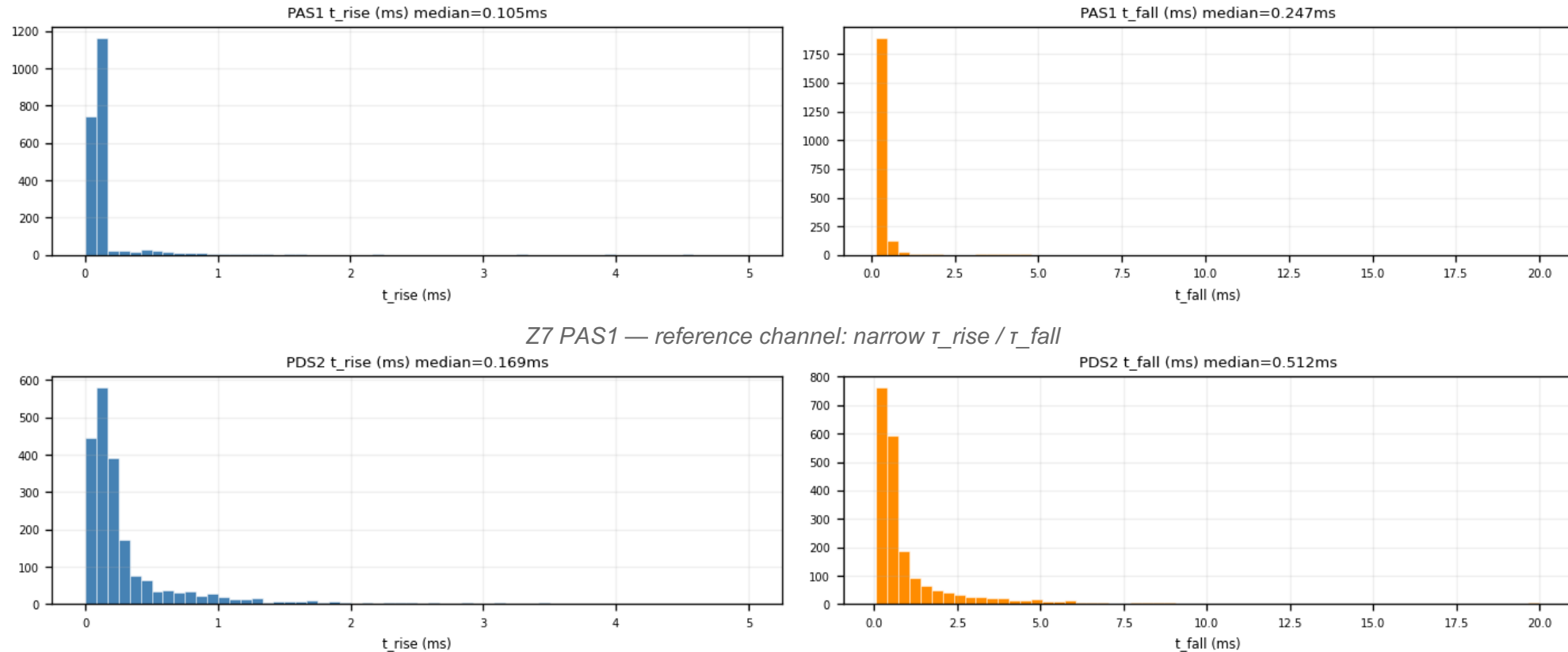
- Selection: well-fit (median NRMSE ≤ 0.4) AND slow (median $\tau_{\text{rise}} > 0.2$ ms) — raw traces show real pulses, not noise
- They are the faint displaced bundle seen in the aligned overlays — real shape variation (candidate surface/bulk effect)
- Exactly the kind of shape variation the multi-template NxM method is built to capture



Z7 shadow events (first 4 channels): raw (gray) / LP (blue) / fit (red) — genuine slow pulses, onset aligned, peak late

Side finding: long- τ_{fall} events on Z7 are a one-channel artifact

- PDS2 shows a broad τ_{fall} tail (median 0.51 ms vs ≈ 0.25 ms elsewhere) and a smeared τ_{rise} distribution
- Cross-channel check: the same events are normal fast pulses in the other 11 channels $\rightarrow$ a PDS2-only low-frequency disturbance, not slow physics

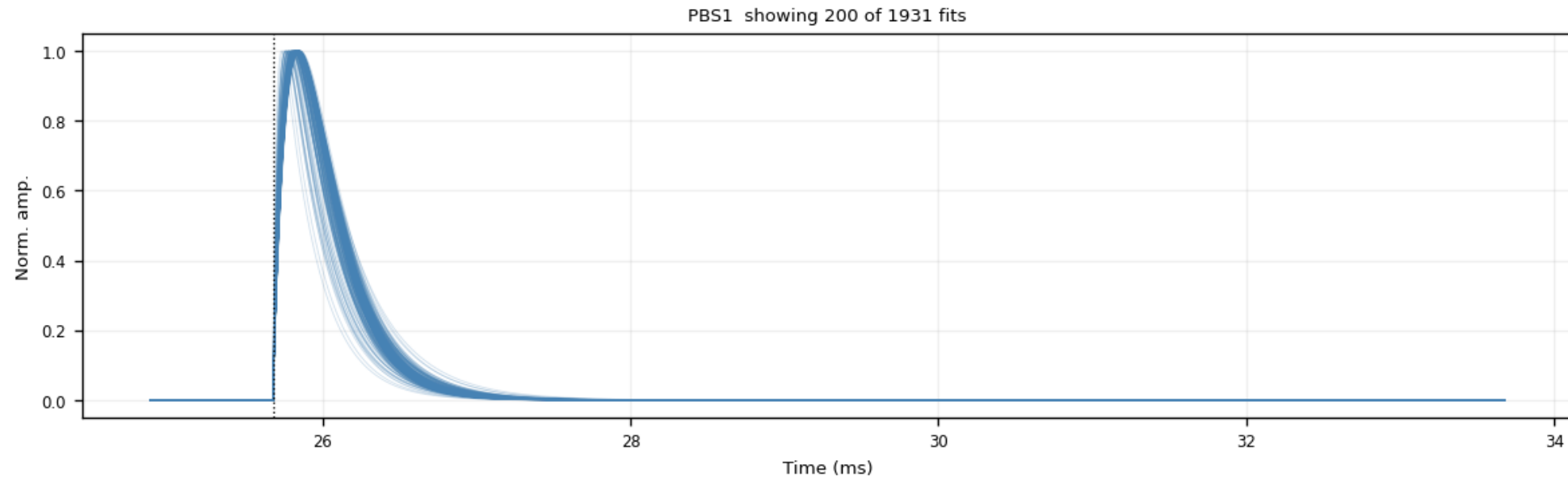


Z7 PAS1 — reference channel: narrow τ_{rise} / τ_{fall}

Z7 PDS2 — broad tails from a channel-specific low-frequency artifact

Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

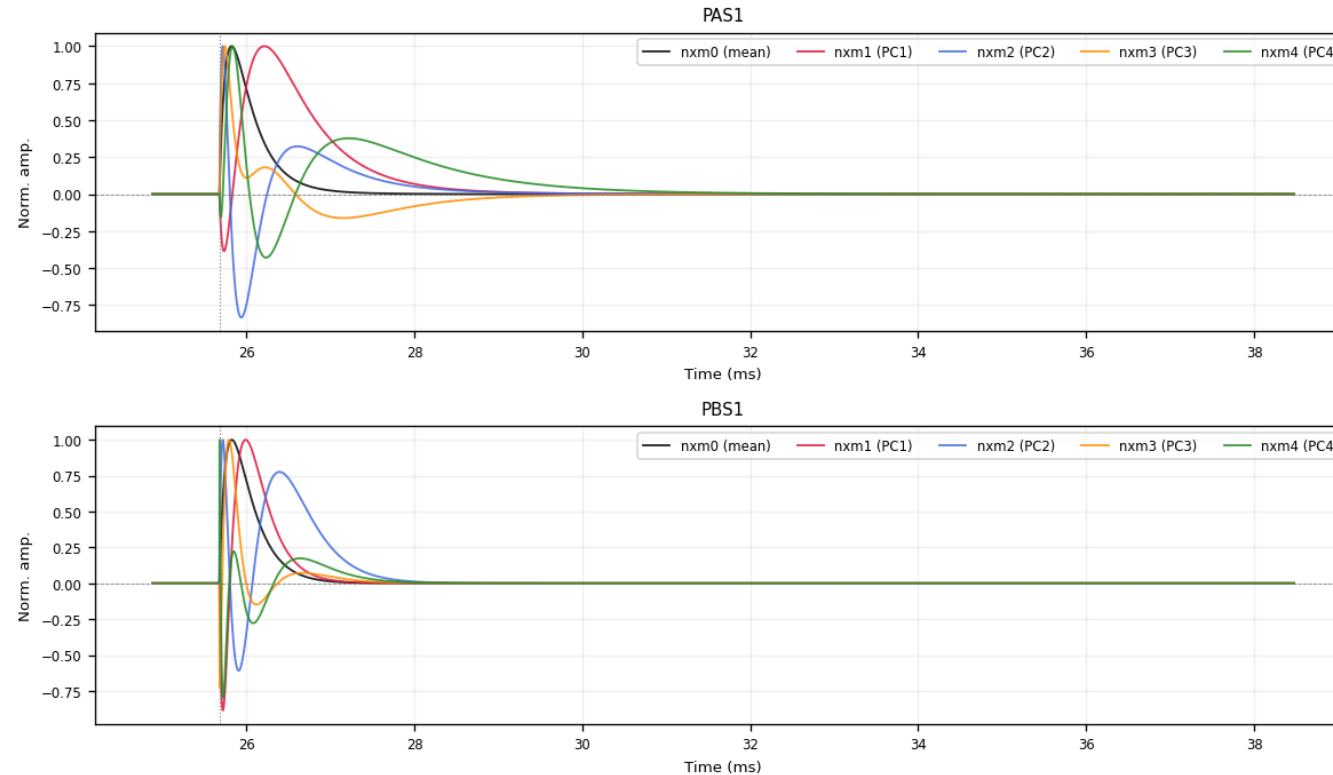
- Per channel: weighted mean of ALL fit_ok fitted curves at the common onset, weight $w = 1/\max(\text{NRMSE}, 0.01)^2$
- Badly-fit events count less but are never excluded — robust and smooth (noise-free by construction)
- Delivered as peak-normalized 32768-bin ROOT TH1D per channel + summed PT/PS1/PS2 templates



Z7 PBS1, fitted curves after $\text{NRMSE} \leq 0.4$ and $\tau_{\text{rise}} \leq 0.3$ ms — the NRMSE-weighted mean of this family is the 1x1 template

Template family 2 — NxM PCA templates

- Input: fitted curves passing $\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$ ms, at common onset, peak-normalized (PCA window 15550–24050, ≤ 3000 curves/channel)
- nxm0 = mean shape; $\text{nxm1} \dots \text{nxm4}$ = first four principal components — a real pulse is fit as $\sum_i \text{amp}_i \cdot \text{nxm}_i$
- PC1 + PC2 already capture 96–98% of the shape variance; all five delivered peak-normalized



Z7 PAS1 / PBS1: nxm0 (mean) + nxm1-4 (PCs) — the oscillating components encode rise/fall-time variation

Delivered — and what remains

- Delivered for all 13 zips, official cdmsbats PulseTemplates layout (zip{N}/{chan}, {chan}nxm0–4, summed PT/PS1/PS2):
 - SNOLAB_R4_20260706_ZhihengLi_zip{N}.root — 2-exp NRMSE-weighted (1x1)
 - SNOLAB_R4_20260707_ZhihengLi_pca_zip{N}.root — NxM PCA (normalized)
- Every step is traceable: raw cache → per-series fit checkpoints indexed by EventNumber → self-documenting figures (11 types × 13 zips, all versioned)
- Cuts derived from the data, validated in raw traces: $\text{NRMSE} \leq 0.4$ (bimodal valley), $\tau_{\text{rise}} \leq 0.3$ ms (drift leakage)
- Open items:
 - final decision on the τ_{rise} ceiling vs the genuine slow-pulse population
 - run the group's template-validation method on the new templates